

19.6 m² room with bathroom

Electrics, lighting, ventilation, sensors, control cabinet

The kit is based on the 3D model (version 54), the owner's sketches and photos. These are working materials for procurement and installation, not a stamped design: the DB and incoming supply diagrams are to be agreed with the management company and the electrical testing laboratory.

Kit contents

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Also in the archive: 3D model (opens in a browser offline), list tables (Excel).

Key decisions

- All cables run on the ceiling in the 18 cm black ceiling band, 90° turns only; drops on walls in black flame-retardant PVC conduit; cable VVGng(A)-LS.
- Lighting: 3 circuits — entry / loft (T0, T3, 3000 K spots) / work (T1, T2, 4000 K linear, ≈500–650 lx); SW2 under the board, push-button, drives impulse relays.
- Ventilation Ø125: supply ≈100 m³/h with G4 filter and DVS EKA 125 1.5 kW heater, two exhausts ≈50 and ≈60 m³/h; Vents TT PRO 125 fans; airflow set by iris dampers; each fan has its own thresholds.
- Automation in 3 levels: 0 — wiring (heater circuit, light buttons); 1 — OWEN PR200 + PRM, hard logic; 2 — AI on a PC, only requests, can be switched off with no consequences.
- The PC may come later and second-hand; all works without it.
- Midea Persona 09 air conditioner (black) — above the control cabinet, AI controls it via Wi-Fi through the PC; two PoE cameras with no blind spots; blue cabinet lighting.
- DB — EKF ShchURn 3/30 two-door 580×490×165, mounted upside down: meter at bottom right, window ≈1.4–1.6 m above floor; conduit entries at the top.

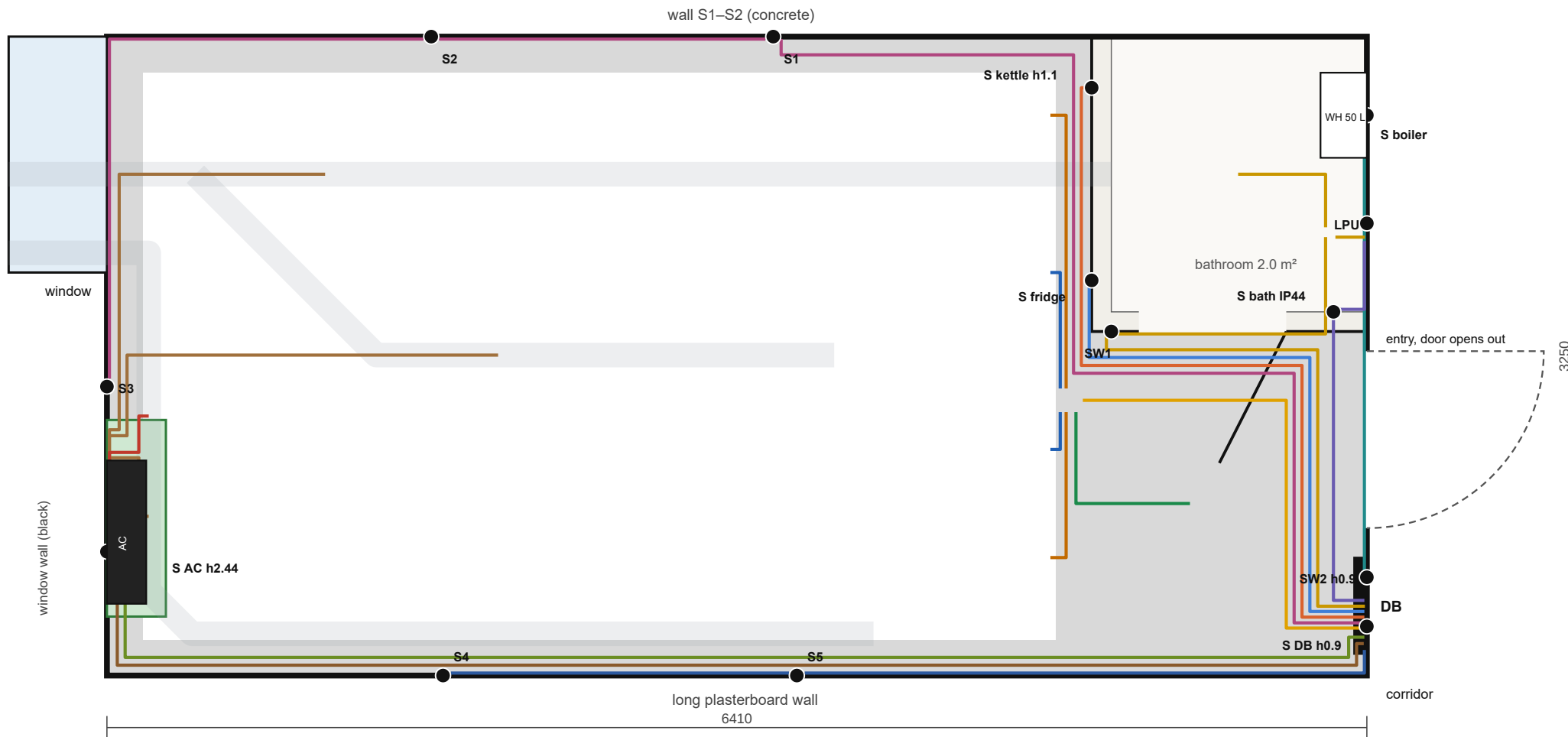
Door opens outward — what is accounted for

- The opening direction matches the evacuation direction — correct from a fire safety standpoint.
- Nothing obstructs the door inside: SW2, the socket under the board and the DB are on the handle side, next to the door casing.
- The open leaf takes ≈0.9 m of the corridor — check that it does not block the passage or neighboring doors; if needed, add a door closer and door stop.
- Hinges outside — door with anti-lift pins.
- Above the door: incoming cable along the ceiling, clearance to the opening ≈74 cm; smoke detector by the entrance on the black ceiling, the leaf does not touch it.
- Corrected on all plans and in the 3D model (door swings into corridor).

To be confirmed on site

- Position of heating pipes (taken from photos ±10–15 cm) — determines the location of S3 and feeds to the cabinet.
- Incoming cable and meter — per management company's conditions.

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General data	Sheet
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- cct 1 — bathroom lighting (SW1, luminaire, backlight), 3×1.5
- cct 2 — room lighting: DB → SW2 and DB → JB1, 5×1.5; JB1 → tracks, 3×1.5
- cct 3 — bathroom socket + LPU, 3×2.5
- cct 4 + 5 — board → control cabinet CC, 2×(3×2.5) in Ø32
- from CC — fans 3×1.5, heater 5×1.5
- cct 6 — boiler, 3×2.5
- cct 7 — fridge

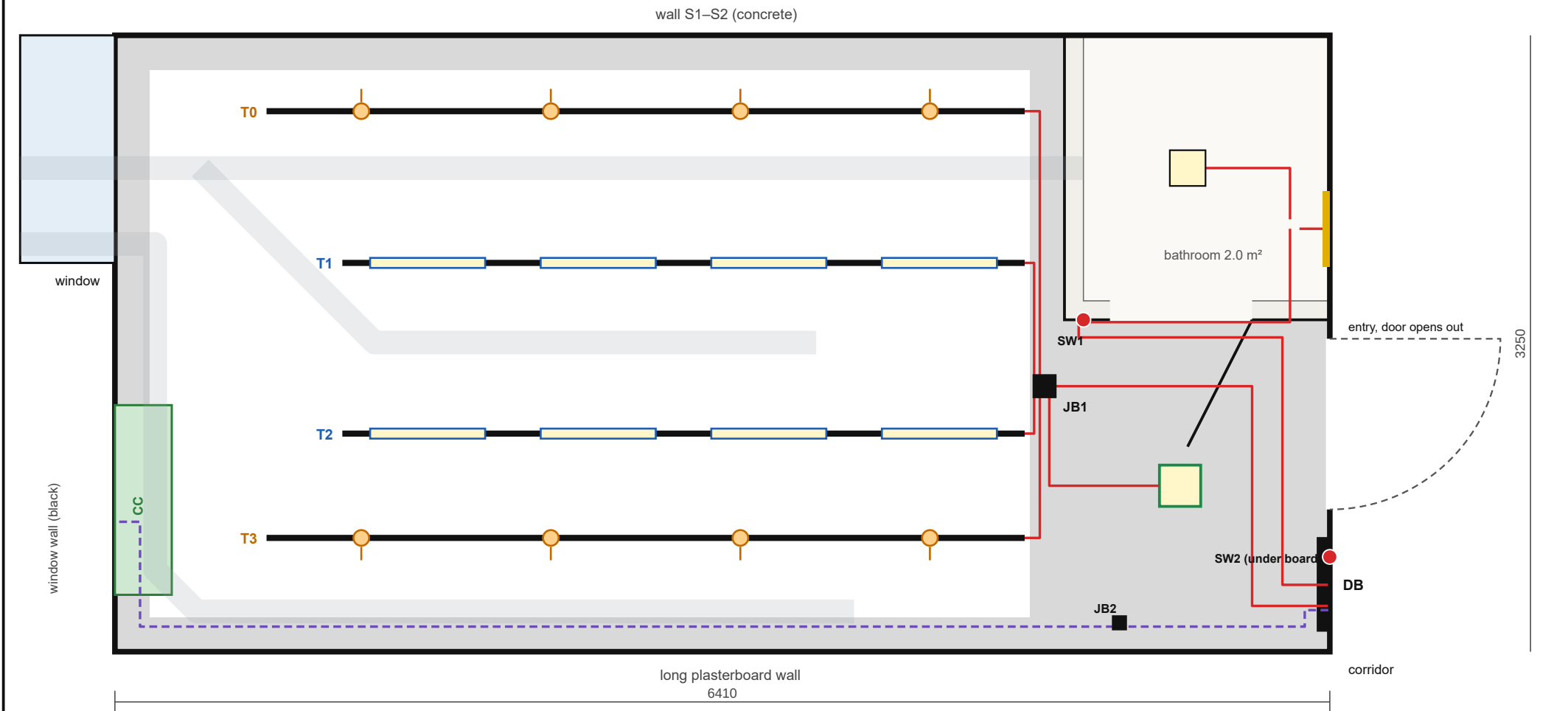
- cct 8 — kettle
- cct 9 — S5 → S4
- cct 10 — S1 → S2 → S3
- cct 11 — AC
- cct 12 — socket under board (spare)

All routes on ceiling in 18 cm black ceiling band, 90° turns only; wall drops in black conduit.

h labels — center height above floor, m. Other sockets — 0.3 m.

Incoming VVGng 5×6 in Ø32 (conduit slid over the laid cable) — along bathroom ceiling to board, no joints.

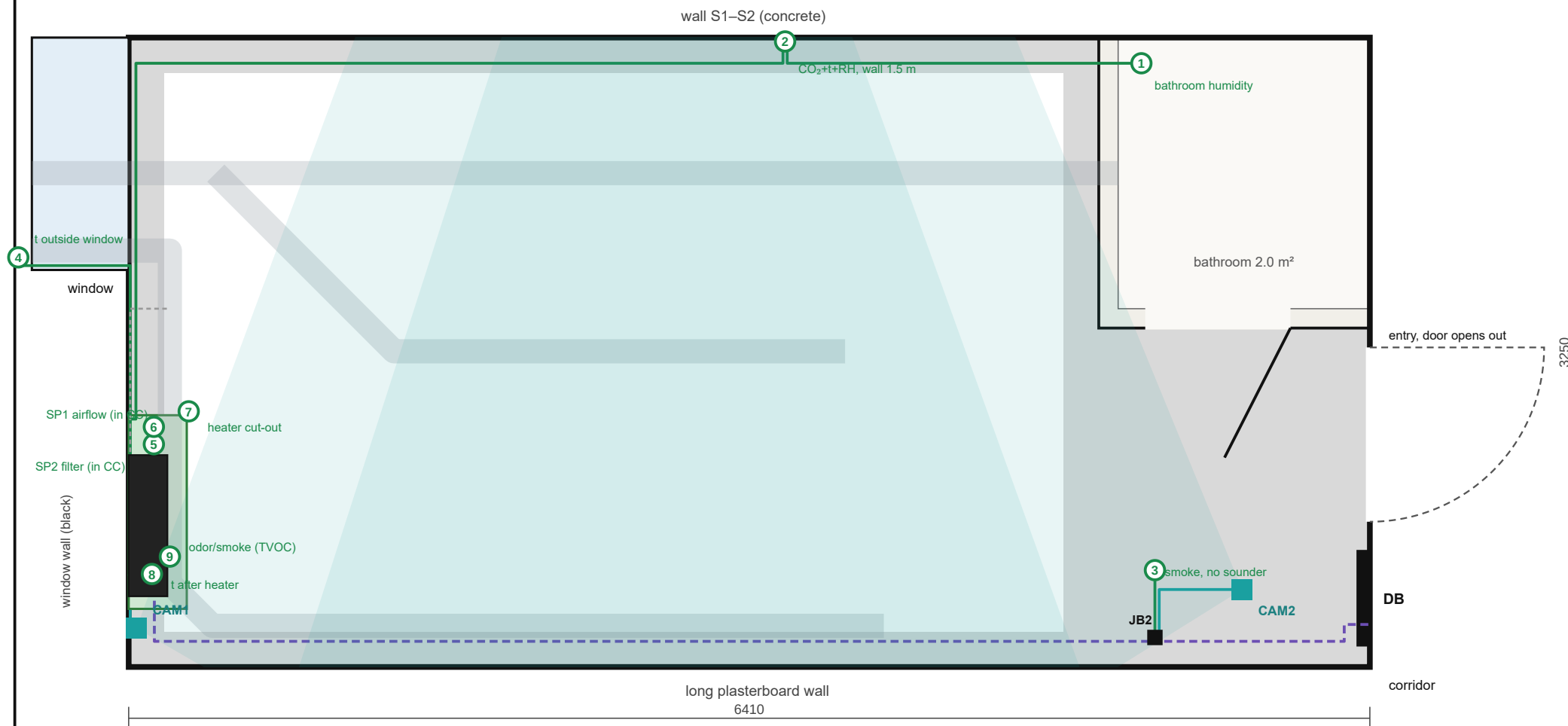
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Power wiring route plan	Sheet
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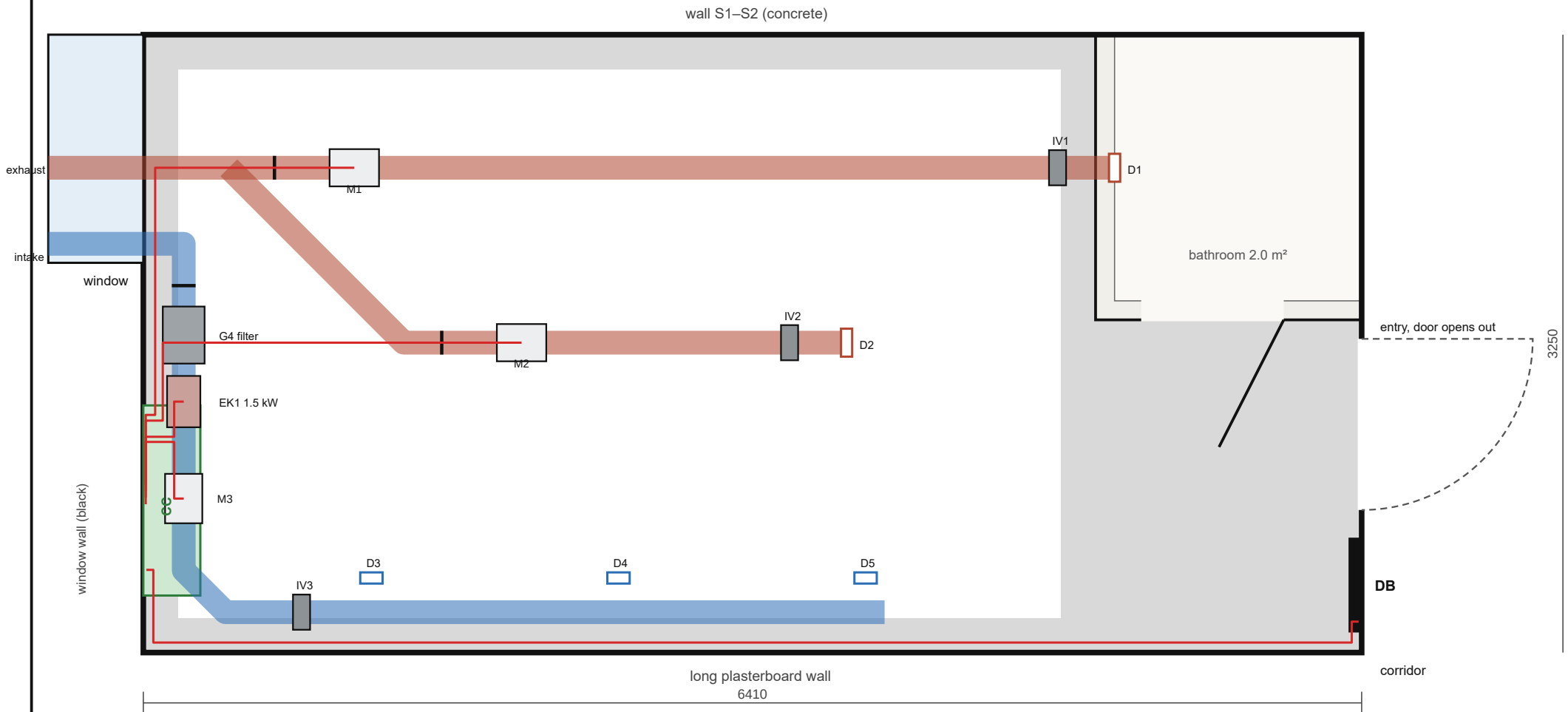
- surface single-phase track 2 m, black (T0–T3)
- linear track luminaire 35 W 4000 K, 607 mm — work (T1, T2)
- track spot 25 W 3000 K, arrow — aim direction — loft (T0, T3)
- surface square 20 W at entry; bathroom luminaire Feron 51920
- lighting cables VVGng(A)-LS 3×1.5 / 5×1.5 in black conduit Ø20–Ø25
- lighting control twisted pair DB ↔ CC (24 V)

SW2 — push-button, 3 momentary keys: 1 — entry, 2 — loft (T0+T3), 3 — work (T1+T2).
Buttons toggle impulse relays KL1–KL3 in the DB directly, not via the CC.
Controller “presses” the same relays via KV1–KV3 (24 V), reads state from their contacts.
JB1 — box on the black ceiling at entry: 5×1.5 from DB (3 lines + N + PE) → 5 lines 3×1.5.
SW1 — 2 keys: bathroom luminaire / mirror backlight (cct 1), one Ø25 conduit to the bathroom box.
Door opens outward — SW2 and socket under board on the handle side, nothing is blocked.

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Lighting plan	Sheet
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Sensors and low-voltage lines plan	Sheet
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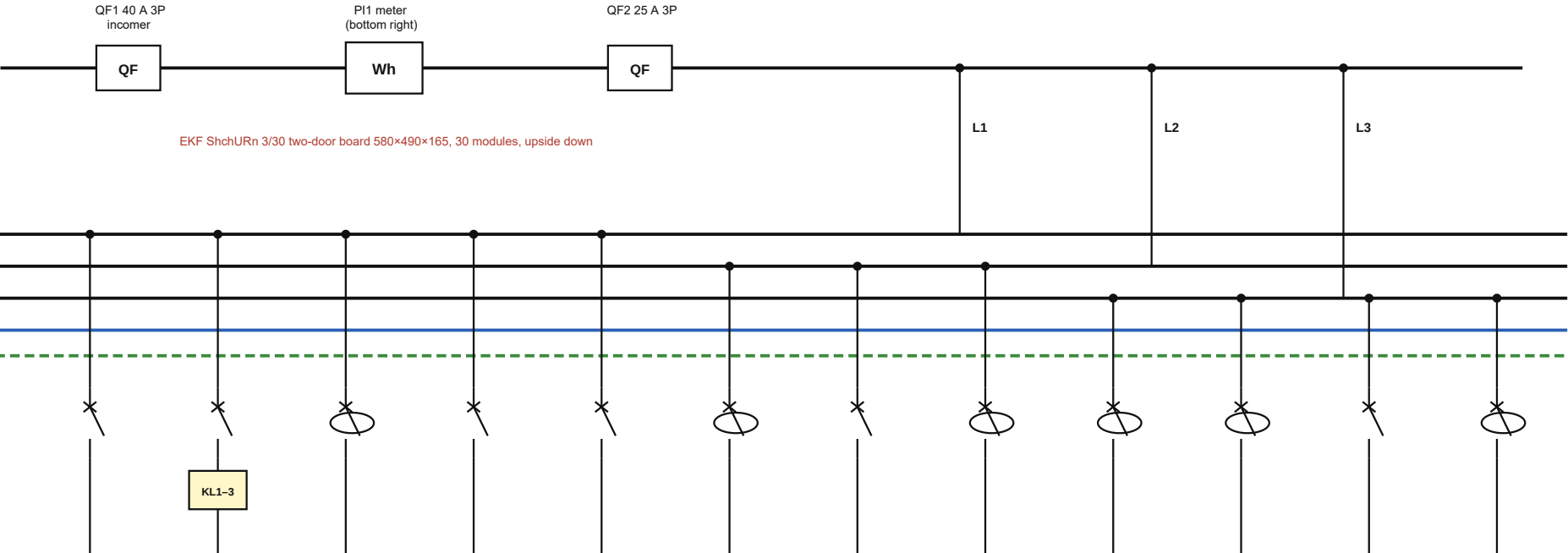


- exhaust Ø125: bathroom ≈50 m³/h (straight) and room ≈60 m³/h (45° bend)
- supply Ø125 ≈100 m³/h: along window to plasterboard wall, 2×45°
- M1–M3 — Vents TT PRO 125; exhausts — low speed, supply — high
- EK1 — DVS EKA 125 1.5 kW (minimum 70 m³/h); IV1–IV3 — iris dampers
- cross bar — spring-loaded backdraft damper
- power from CC: fan and heater terminals face the ceiling

Blank in left sash: ALAV 125 grilles 40 cm apart, splitter plate between. Exhaust outlet — fins sideways, to wall S1–S2; intake — fins down.
 Ceiling-to-duct gap 15 cm; supply 15 cm off window wall and plasterboard wall.
 D1–D5 — Rodfer DK-125 ChM diffusers (black, Ø180, adjustable): D1 in bathroom wall, D2 end-mounted, D3–D5 supply.
 Supply order: damper → filter → heater → fan (fan receives warm air). Flexible connectors VVG 125 — 2 per fan.
 Airflow is set once with iris dampers: IV1 ≈50, IV2 ≈60, IV3 ≈100 m³/h (not below 70 — heater minimum).
 Independent operation: each fan has its own controller relay and its own switch-on thresholds.
 Heater — only with supply running and airflow present (switch 6), see CC diagram.

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Ventilation plan	Sheet
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From MSB
3~ 380/220 V
L1 L2 L3 N PE
VVGng 5×6



Circuit	1	2	3	4	5	6	7	8	9	10	11	12
Phase	L1	L1	L1	L1	L1	L2	L2	L2	L3	L3	L3	L3
Device	QF6 6 A, C	QF7 6 A, C	QFD2 RCBO 10 A, 30 mA	QF9 10 A, C	QF10 10 A, C	QFD1 RCBO 16 A, 30 mA	QF11 6 A, C	QFD3 RCBO 16 A, 30 mA	QFD4 RCBO 16 A, 30 mA	QFD5 RCBO 16 A, 30 mA	QF15 10 A, C	QFD6 RCBO 16 A, 30 mA
Cable VVGng(A)-LS	3×1.5	5×1.5 3×1.5	3×2.5	3×2.5	3×2.5	3×2.5	3×2.5	3×2.5	3×2.5	3×2.5	3×2.5	3×2.5
Length w/ allowance, m	16.0	11.0	6.3	9.7	9.7	4.0	7.0	7.3	13.9	25.2	8.8	1.1
Pinst, kW	0.05	0.5	2	1.5	0.6	2	0.3	2.2	2	2.5	1	1
Load	Bathroom light, backlight (SW1)	Room light: entry, loft, work (SW2)	Bathroom socket, LPU	CC: supply heater	CC: fans, PC, UPS, 24 V	Water heater 2 kW	Fridge	Kettle	Sockets S4, S5	Sockets S1, S2, S3	A/C	Socket under board (spare)

Changes from the first revision of the diagram:

- 30-module board: phase MCBs QF3–QF5 removed (QF2 25 A does their job), RCBOs in a 1-module (18 mm) housing;
- cct 3 (bathroom), 8, 9, 10, 12 — 30 mA RCBOs (PUE 7.1.79, 7.1.82–7.1.85); 100 mA fire-protection RCD optional, in a separate box;
- cct 2 — via impulse relays KL1–KL3 and 24 V relays KV1–KV3 (sheet "Lighting control diagram");
- cct 4, 5 — to control cabinet CC instead of the ventilation control board; cct 5 — 10 A (PC, UPS, fans, 24 V PSU);
- cct 12 — new: double socket under the board; cct 1 — 2-key SW1 (light / mirror backlight).

Installed power per phase: L1 ≈ 4.6 kW, L2 ≈ 4.5 kW, L3 ≈ 6.5 kW (without diversity factor).

19.6 m² room with bathroom — electrics, lighting, ventilation, automation

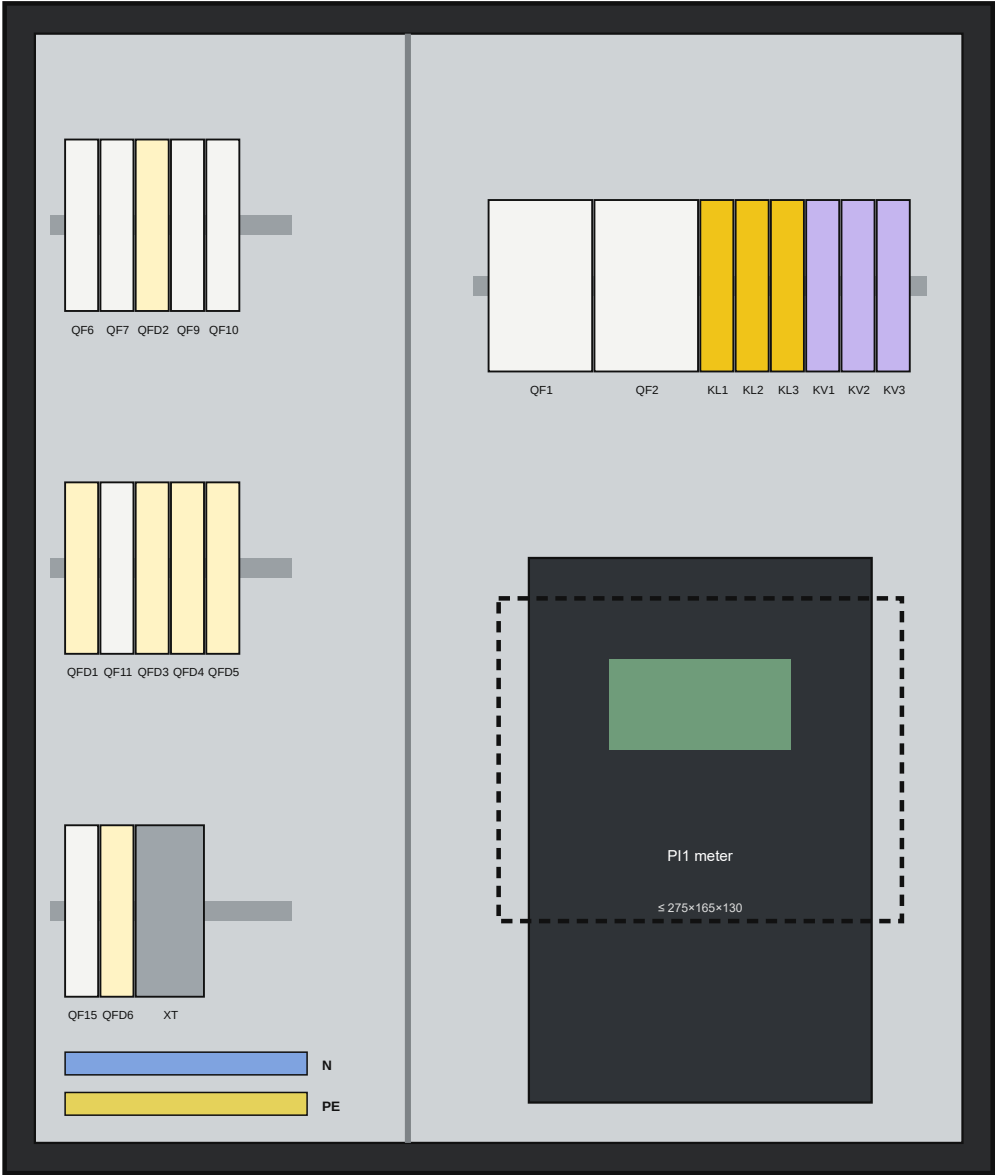
DB single-line diagram

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DB EKF ShchURn 3/30 two-door 580×490×165, upside down; front view, doors removed, scale 1:3.75



conduit entries at the top (after inverting, the factory Ø30 holes are at the top); drill extra for ~11 conduits
board bottom at 1.30 m; meter window (dashed) =1.4–1.6 m above floor — readable at 1.55 m height
under the board: SW2 and 2-gang socket at 0.9 m, fed from the board bottom

DB device list

QF1	Incoming MCB 40 A, 3P, C (before meter)	1
PI1	3-phase meter (per conn. conditions), metering bay bottom right	1
QF2	MCB 25 A, 3P, C (after meter)	1
QF6, QF7	MCB 6 A, 1P, C — lighting	2
QF9, QF10, QF15	MCB 10 A, 1P, C — CC ×2, air conditioner	3
QF11	MCB 6 A, 1P, C — fridge	1
QFD1, QFD3–QFD6	RCBO 16 A 30 mA type A, 1P+N, 1 module	5
QFD2	RCBO 10 A 30 mA type A, 1P+N, 1 module — bathroom	1
KL1–KL3	Impulse relay 16 A, 230 V, with status contact	3
KV1–KV3	DIN interface relay, 24 V DC coil	3
XT	DIN terminals for control twisted pair (8 cores)	1
—	N, PE busbars — supplied with board; comb busbars/links, blanks	1

26 of 30 modules used, 4 free (1 in each column row and 2 in the bottom one).
Board is upside down: devices mounted as usual — “ON” handle up;
re-stick labels and door diagram. Metering bay is sealed by the grid operator.
100 mA fire-protection RCD is not in the 30 modules — if wanted,
it goes in a separate 4–6 module box next to the board.

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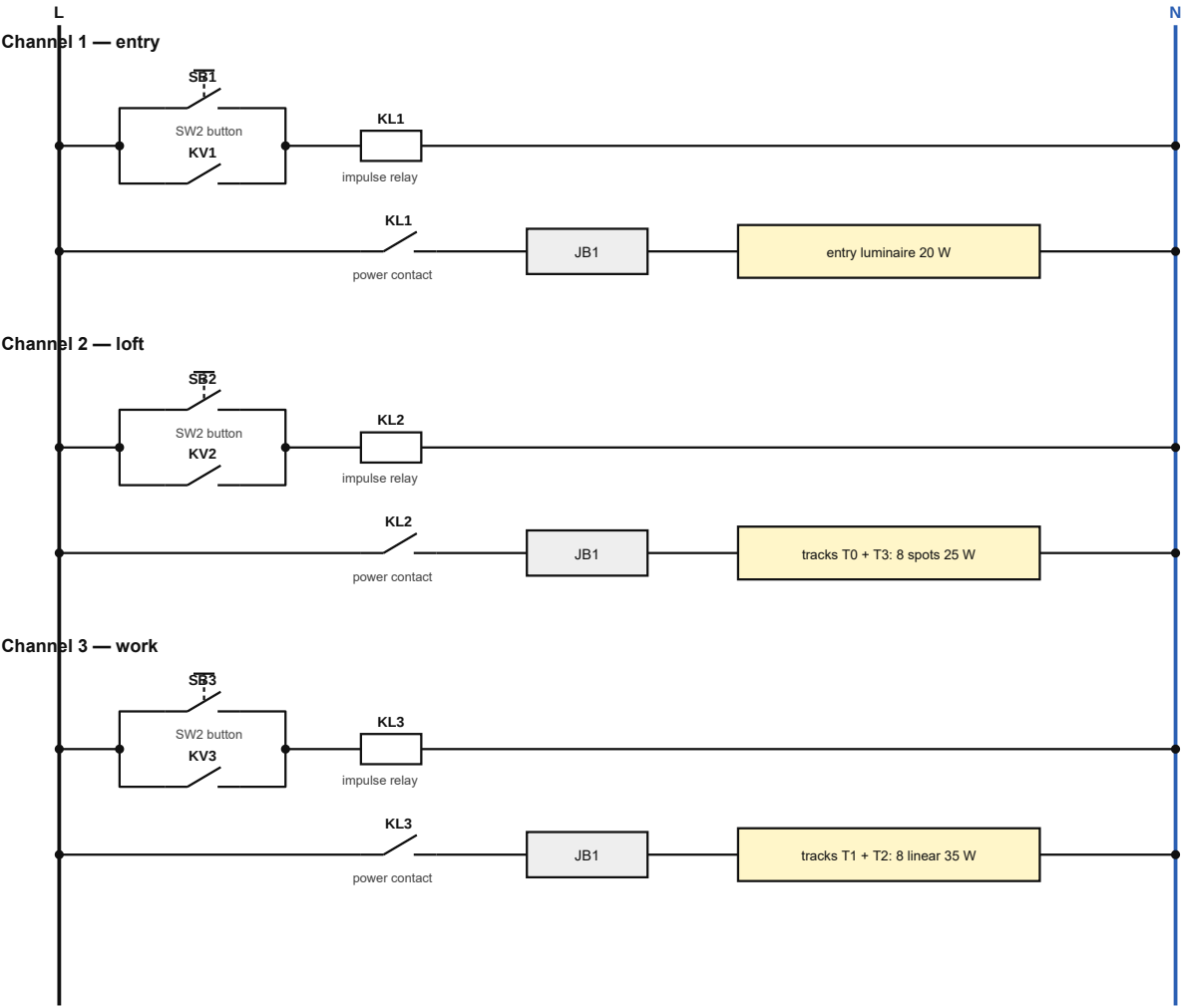
DB layout

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230 V circuits (cct 2, QF7 in DB)



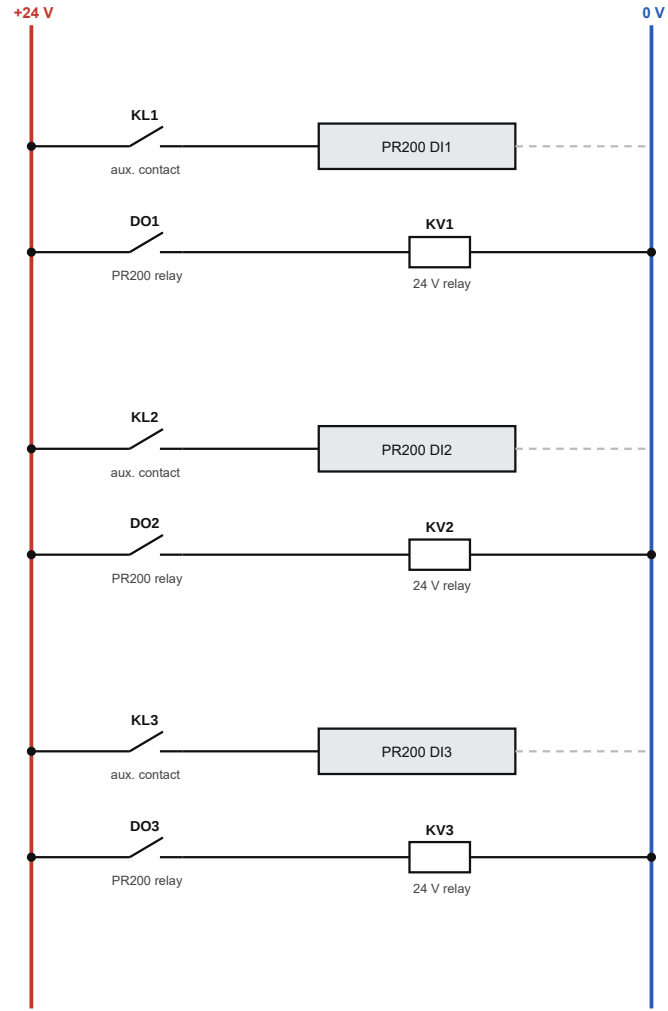
Buttons SB1–SB3 (SW2) and contacts KV1–KV3 are in parallel: a short press on any toggles impulse relay KLn.

Without the control cabinet (controller off, no 24 V) the buttons work as usual. The controller “presses” the relay with a 0.2–0.5 s pulse.

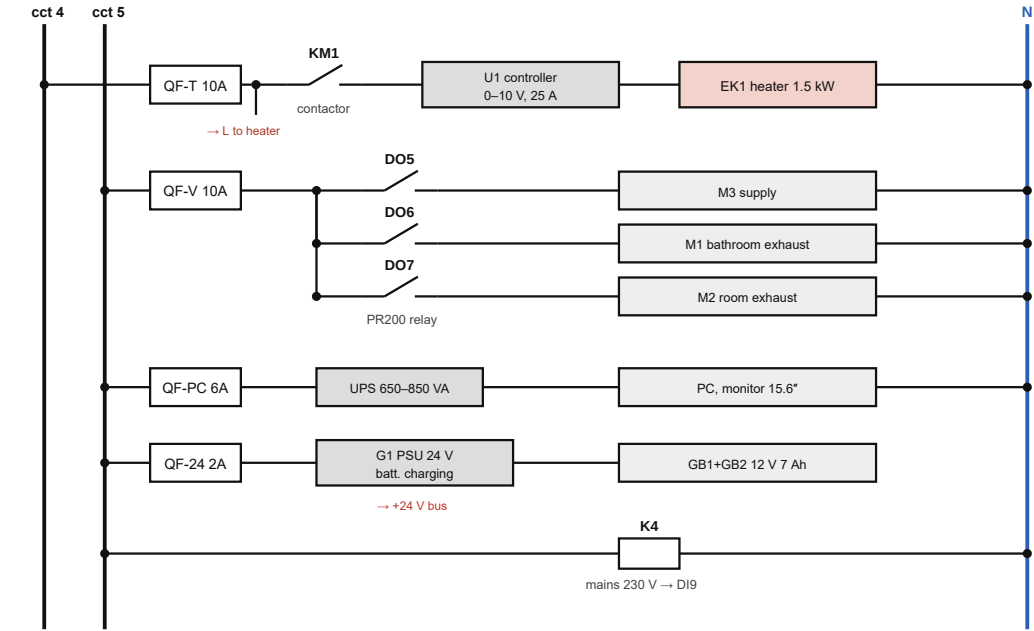
The controller reads each circuit’s state via the KLn aux. contact — AI and automation always know whether the light is on.

Twisted pair DB ↔ CC (8 cores): +24 V, 0 V, DO1–DO3 → KV1–KV3, KL1–KL3 → DI1–DI3. KL1–KL3, KV1–KV3 are in DB.

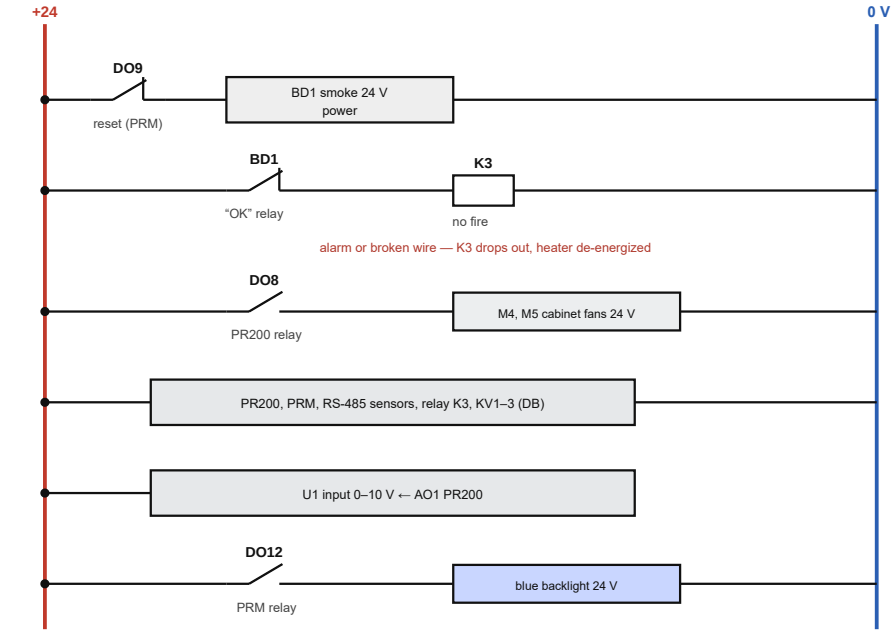
24 V circuits (from CC via twisted pair)



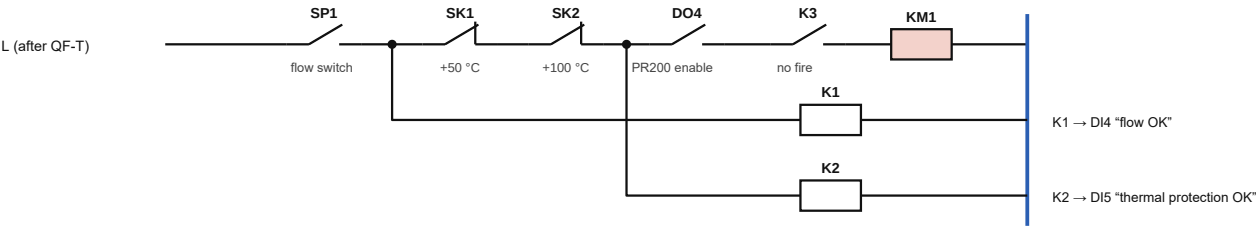
230 V power circuits



24 V circuits



Heater enable chain (230 V) — works without the controller



Main principle: the heater can turn on only if all links of the KM1 coil chain are closed at once:
air flow present (SP1), both thermal cut-outs intact (SK1, SK2), controller enables (DO4), no fire signal (K3).
Links SP1, SK1, SK2, K3 are hardware: they work with the controller and computer hung or off.
U1 sets power smoothly (0–10 V from AO1) but sits after KM1 — without KM1 no current reaches the heater.
Heater shutdown: first AO1 = 0, then DO4; supply fan M3 runs another 2–3 min to cool the heater (PR200 program).

OWEN PR200-24.4.2.0 (DI1–8, AI1–4, AO1–2, DO1–8, 2×RS-485) + PRM-24.1 (DI9–16, DO9–16). 24 V supply from G1 with battery.

Channel	What	From / to
DI1	"Entry" light on	KL1 aux. contact (DB)
DI2	"Loft" light on	KL2 aux. contact
DI3	"Work" light on	KL3 aux. contact
DI4	Air flow present	K1
DI5	Heater thermal prot. OK	K2
DI6	Fire	BD1 / K3
DI7	Heater contactor on	KM1 aux. contact
DI8	Filter clogged	SP2 diff. press.
DI9	Mains 230 V OK	K4
DI10	CC door button	button on side panel
DI11	CC door closed	reed sw. SQ1
DI12–16	Spare	—
AI1–AI4	Spare (temperatures now via RS-485)	—
AO1	Heater power 0–10 V	U1 controller
AO2	Spare	—

Channel	What	From / to
DO1–DO3	"Press" light: entry / loft / work	KV1–KV3 (DB)
DO4	Heater enable	KM1 circuit
DO5	Supply fan	M3
DO6	Bathroom exhaust	M1
DO7	Room exhaust	M2
DO8	Cabinet fans 24 V	M4, M5
DO9	Smoke detector reset	BD1 power
DO10–DO11	Computer Power / Reset buttons	PC mobo
DO12	Cabinet blue backlight	24 V strip
DO13	CC door: open	reverse relay → drive
DO14	CC door: close	reverse relay → drive
DO15–16	Spare	—
RS-485 No. 1	Sensor bus, PR200 is master	4 iArduino sensors + R4DCB08
RS-485 No. 2	Link to computer, PR200 is slave	USB–RS-485 in PC
PC (not PR200)	A/C, cameras CAM1–CAM2	Midea Wi-Fi (local), PoE switch

AI link: the computer writes setpoints and commands to PR200 registers, plus a heartbeat counter every 5 s. No update for 30 s — PR200 drops AI commands and runs on its own setpoints.

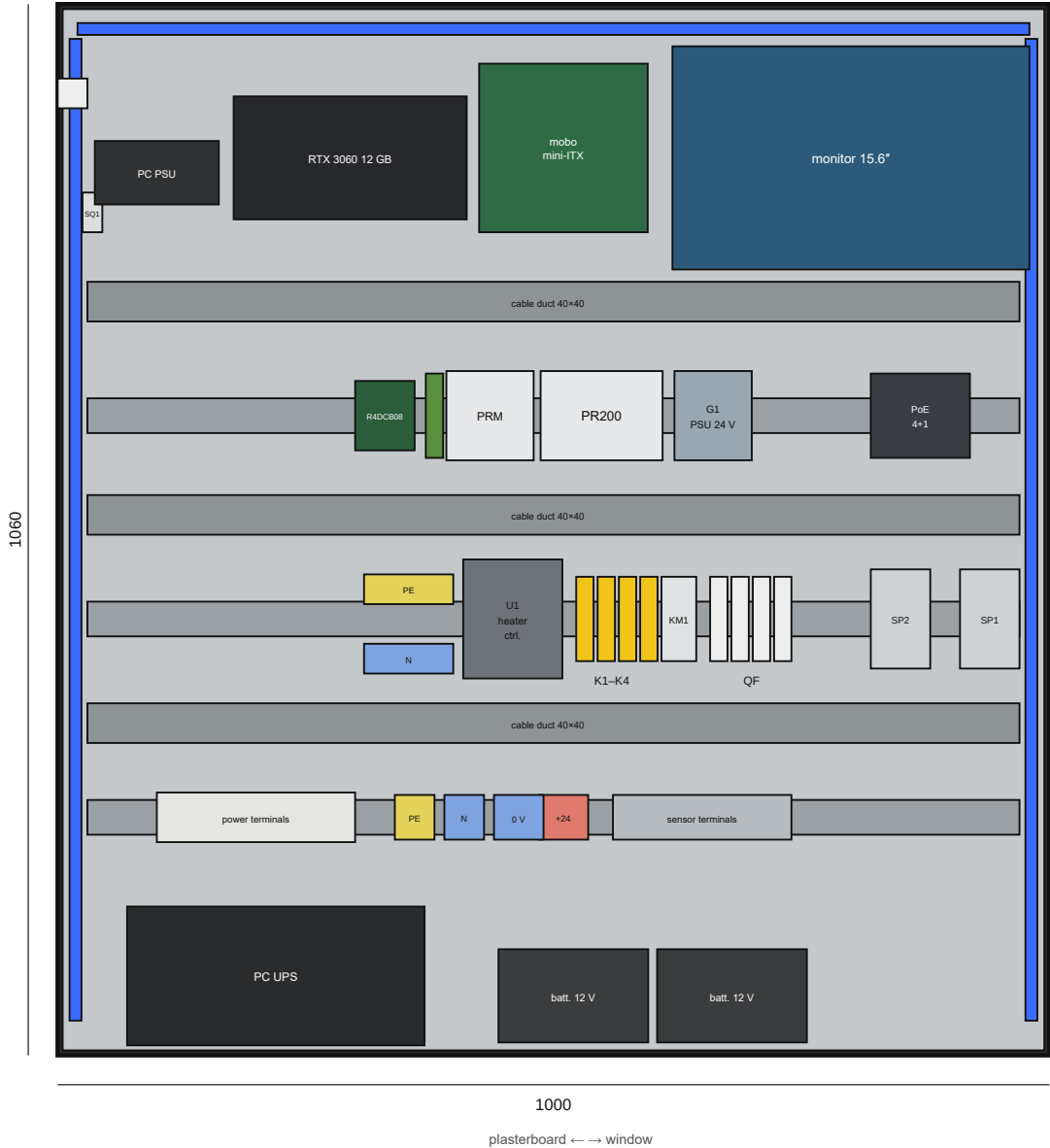
Limits the AI cannot exceed (example): supply t 14...24 °C, bathroom humidity on 55...80 %, CO₂ on 700...1200 ppm, heater ≤ 100 %; lighting — unrestricted.

Example “standard mode” setpoints (changed from the PR200 screen):

- bathroom exhaust: on RH > 70 %, off RH < 60 % + 5 min; • room exhaust and supply: on CO₂ > 900 ppm, off < 700 ppm;
- heater: only with supply running and outdoor t < +15 °C, holds supply t at +18 °C; • cabinet fans: cabinet t > 35 °C;
- fire: stop heater, stop ventilation, signal to PC; • mains loss: run on battery, PC shuts down from its own UPS;
- odor/smoke in supply air (sharp TVOC rise): stop supply and both exhausts for 10–15 min, then supply for 1 min and re-check;
- CC door: button — move / stop / reverse, travel no longer than 35 s (else stop and alarm); open > 5 min — notification.

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CC: controller inputs and outputs	Sheet
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CC — front view through the transparent door, scale 1:7.4 (cabinet bottom 0.76 m above floor)



Welded steel enclosure, continuous welds, 1000 × 1060 × 300 mm, bottom at 0.76 m.

Door: 2 tempered glass panes 4 mm, 540 mm each, lowering below the cabinet on guides (posts extended down by 600 mm); 24 V actuator, 300 N, 2:1 pulley.

Heating pipes above (≈2.6–2.7 m): roof welded with a continuous seam, no openings; no canopy — drips from the pipes cannot get inside.

All conduits with wires enter via IP68 cable glands in the roof; drip loop before each entry.

Cabinet ventilation: intake from below through a filtered grille in the floor, exhaust by 2 × 24 V fans in the side wall on the plasterboard side (DO8, by cabinet t).

Zones: bottom — UPS and battery; terminals; power rail; control rail; top — computer and monitor (heat rises).

The computer can be added later — automation works without it.

The computer can be second-hand; UPS and batteries — new.

DIN rails 35 mm, at: 0.24 / 0.44 / 0.645 m from cabinet bottom.

Blue backlight — 24 V strip around the perimeter (DO12).

SP1, SP2 — diff. pressure switches, tubes to supply duct via top cable gland.

R4DCB08 — temperature module (DS18B20 → RS-485); SQ1 — door reed switch.

Air conditioner above the cabinet (15 cm clearance to cabinet roof).

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Control cabinet CC layout	Sheet
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No.	Line	From → to	Cable	Conduit	Route, m	Cable incl. allowance, m
1	Cct 6 — boiler	DB → boiler socket	VVGng(A)-LS 3×2.5	Ø25	3.32	4.0
2	Cct 3 — bathroom socket	DB → double socket	VVGng(A)-LS 3×2.5	Ø25	4.10	4.8
3	Cct 3 — loop to LPU	double socket → LPU	VVGng(A)-LS 3×2.5	Ø25	1.05	1.5
4	Cct 1 — SW1 supply	DB → SW1	VVGng(A)-LS 3×1.5	Ø20	5.34	6.0
5	SW1 → light and backlight (one conduit)	SW1 → junction box	2×(3×1.5)	Ø25	3.35	7.7
6	Box → luminaire	box → luminaire	VVGng(A)-LS 3×1.5	Ø20	0.68	cable continues from shared conduit
7	Box → backlight	box → backlight	VVGng(A)-LS 3×1.5	Ø20	1.21	cable continues from shared conduit
8	Cct 7 — fridge	DB → fridge socket	VVGng(A)-LS 3×2.5	Ø25	6.28	7.0
9	Cct 8 — kettle	DB → kettle socket	VVGng(A)-LS 3×2.5	Ø25	6.59	7.3
10	Cct 9 — DB → S5	DB → S5	VVGng(A)-LS 3×2.5	Ø25	6.24	6.9
11	Cct 9 — S5 → S4	S5 → S4	VVGng(A)-LS 3×2.5	Ø25	6.55	6.9
12	Cct 11 — air conditioner	DB → AC socket	VVGng(A)-LS 3×2.5	Ø25	8.13	8.8
13	Cct 10 — DB → S1	DB → S1	VVGng(A)-LS 3×2.5	Ø25	9.11	9.8
14	Cct 10 — S1 → S2	S1 → S2	VVGng(A)-LS 3×2.5	Ø25	6.49	6.9
15	Cct 10 — S2 → S3	S2 → S3	VVGng(A)-LS 3×2.5	Ø25	8.12	8.5
16	Spare — socket under DB	DB → socket under DB	VVGng(A)-LS 3×2.5	Ø25	0.36	1.1
17	Cct 4 + cct 5 → CC	DB → CC	2×(3×2.5)	Ø32	8.66	19.3
18	CC → bathroom exhaust fan	CC → fan	VVGng(A)-LS 3×1.5	Ø25	3.67	4.0
19	CC → room exhaust fan	CC → fan	VVGng(A)-LS 3×1.5	Ø25	3.65	4.0
20	CC → supply fan	CC → fan	VVGng(A)-LS 3×1.5	Ø25	1.72	2.0
21	CC → heater 1.5 kW	CC → heater (2 cores — thermal cut-out)	VVGng(A)-LS 5×1.5	Ø25	1.61	1.9
22	Cct 2 — DB → SW2	DB → SW2	VVGng(A)-LS 5×1.5	Ø20	0.36	1.1
23	Cct 2 — DB → box JB1 (3 keys)	DB → JB1	VVGng(A)-LS 5×1.5	Ø20	3.41	4.1
24	JB1 → track T0 (loft)	JB1 → T0	VVGng(A)-LS 3×1.5	Ø20	1.45	1.8
25	JB1 → track T1 (work)	JB1 → T1	VVGng(A)-LS 3×1.5	Ø20	0.63	1.0
26	JB1 → track T2 (work)	JB1 → T2	VVGng(A)-LS 3×1.5	Ø20	0.23	0.6
27	JB1 → track T3 (loft)	JB1 → T3	VVGng(A)-LS 3×1.5	Ø20	0.80	1.2
28	JB1 → entry luminaire	JB1 → entry luminaire	VVGng(A)-LS 3×1.5	Ø20	1.01	1.4
29	Bus: bathroom humidity sensor → sensor 2	sensor 1 → sensor 2	twisted pair FTP cat5e	Ø16	3.18	3.8
30	Bus: sensor 2 → CC	sensor 2 → CC	twisted pair FTP cat5e	Ø16	7.76	8.1
31	Smoke detector → CC	smoke det. → CC	twisted pair FTP cat5e	Ø16	0.28	0.7
32	Light control: DB → box at entrance	DB → box JB2	FTP cat5e (control)	Ø16	2.01	2.6
33	Smoke det. + light control + camera 2 → cabinet (one conduit)	box JB2 → cabinet	3×FTP	Ø20	6.71	22.2
34	Outdoor temperature → cabinet	outdoor probe → cabinet (R4DCB08)	twisted pair FTP cat5e	Ø16	2.32	2.6
35	t after heater and odor sensor → drop	supply probe → CC	2×FTP	Ø16	0.08	1.0
36	Supply sensors (t after heater, odor) → cabinet	supply duct → CC	2×FTP	Ø20	0.77	2.5
37	Diff. pressure switch tubes (4 pcs) → cabinet	supply → SP1, SP2 in cabinet	PVC pneumatic tube Ø6	Ø16	1.67	cable continues from shared conduit
38	Camera 1 → cabinet (side)	camera 1 → cabinet, side wall	twisted pair FTP cat5e	Ø16	1.12	1.9
39	Camera 2 → box JB2	camera 2 → box JB2	twisted pair FTP cat5e	Ø16	0.57	1.0

VVGng(A)-LS 3×2.5: **92.9 m** VVGng(A)-LS 3×1.5: **31.8 m** VVGng(A)-LS 5×1.5: **7.0 m** twisted pair FTP cat5e: **21.5 m** FTP cat5e (control): **24.8 m** PVC pneumatic tube Ø6: **1.9 m**
conduit Ø16: **19.0 m** conduit Ø20: **22.6 m** conduit Ø25: **80.3 m** conduit Ø32: **12.2 m** incoming cable VVGng 5×6 in Ø32 — 3.6 m in the room (part of Ø32; conduit slid over already laid cable)
Cable allowance: 0.5 m per board, 0.2–0.5 m per point. Route lengths per 3D model, along ceiling with 90° turns.

19.6 m² room with bathroom — electrics, lighting, ventilation, automation

Cable schedule

Sheet

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Item	Qty	Unit	Price, \$	Sum, \$	Price basis
DB board					
MCB 6 A 1P, C (QF6, QF7, QF11)	3	pcs	4.19*	13*	estimate
MCB 10 A 1P, C (QF9, QF10, QF15)	3	pcs	4.19*	13*	estimate
RCBO 16 A 30 mA type A, 1 module (QFD1, QFD3–QFD6)	5	pcs	35.90*	180*	estimate
RCBO 10 A 30 mA type A, 1 module (QFD2, bathroom)	1	pcs	35.90*	36*	estimate
Impulse relay 16 A, 230 V, with status contact (KL1–KL3)	3	pcs	29.92*	90*	estimate
Interface relay, DIN, 24 V coil (KV1–KV3)	3	pcs	8.38*	25*	estimate
Comb busbars/jumpers, terminals, blanks (N/PE busbars come with board)	1	set	29.92*	30*	estimate
Section subtotal (excl. “—” items)				387	
Sockets and switches (black)					
Socket outlet, 4-gang (S1–S5)	5	pcs	17.95*	90*	estimate
Single socket: AC, kettle, fridge, boiler (IP44)	4	pcs	4.79*	19*	estimate
Double socket IP44 in bathroom	1	pcs	14.36*	14*	estimate
Double socket under DB	1	pcs	9.57*	10*	estimate
Two-key switch SW1 (bathroom)	1	pcs	7.18*	7*	estimate
Push-button switch, 3 buttons, momentary, SW2	1	pcs	17.95*	18*	estimate
Section subtotal (excl. “—” items)				158	
Lighting					
Surface-mounted single-phase track 2 m, black	8	pcs	16.54	132	estimate
Ritter Artline linear 35 W 4000 K, 607 mm, black	8	pcs	33.96	272	estimate
Volpe ULB-Q276 25 W 3000 K, track spot	8	pcs	9.93	79	estimate
Smartbuy 20 W IP65, black square (entry)	1	pcs	16.97	17	estimate
Feron 51920 24 W IP44, black square (bathroom)	1	pcs	42.09	42	estimate
Above-mirror light IP44 — to be chosen	1	pcs	—	—	TBC
Power feeds, end caps, track connectors	4	set	4.79*	19*	estimate
Boxes JB1, JB2, bathroom junction box	3	pcs	3.59*	11*	estimate
Section subtotal (excl. “—” items)				572	

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Item	Quan- tity	Unit	Price, \$	Sum, \$	Price basis
Cables and conduit					
Cable VVGng(A)-LS 3×2.5	95	m	1.32*	125*	estimate
Cable VVGng(A)-LS 3×1.5	35	m	0.90*	32*	estimate
Cable VVGng(A)-LS 5×1.5 (light: DB → SW2, DB → JB1; heater + thermal cut-out)	8	m	1.44*	12*	estimate
Twisted pair FTP cat5e (sensors, control)	50	m	0.48*	24*	estimate
Black PVC conduit Ø16 w/ draw wire (Ural-Pak)	20	m	0.22*	4*	estimate
Black PVC conduit Ø20 w/ draw wire (Ural-Pak)	25	m	0.26*	6*	estimate
Black PVC conduit Ø25 w/ draw wire (Ural-Pak)	100	m	0.34*	34*	estimate
Black PVC conduit Ø32 w/ draw wire (Ural-Pak)	15	m	0.48*	7*	estimate
Conduit clips Ø16/20/25/32 (35 cm spacing)	440	pcs	0.07*	31*	estimate
Sleeves in bathroom wall, non-combustible sealant	4	pcs	1.80*	7*	estimate
Section subtotal (excl. “—” items)				282	
Ventilation					
Duct fan Vents TT PRO 125 (2 speeds, 240/350 m³/h, 110/150 Pa, 25/30 W)	3	pcs	89.64	269	estimate
Duct heater DVS EKA 125 1ph 1.5 kW (thermal cut-out +50 °C auto / +100 °C manual)	1	pcs	72.28	72	estimate
Duct filter Ø125 with G4 cassette + spare cassette	1	set	29.92*	30*	estimate
Spring-loaded backdraft damper Ø125	3	pcs	7.18*	22*	estimate
Iris damper Ø125 (airflow adjustment)	3	pcs	15.56*	47*	estimate
Flexible connector Vents VVG 125 (vibration isolation, 2 per fan)	6	pcs	7.18*	43*	estimate
Outdoor grille ALAV 125 (aluminum, angled louvers, mesh)	2	pcs	4.55	9	estimate
Splitter plate between grilles: composite panel 3–4 mm 180×320, black	1	pcs	5.98*	6*	estimate
Diffuser Rodfer DK-125 ChM (steel, matte black, Ø180, adjustable, magnetic mount)	5	pcs	19.99	100	estimate
Ducts Ø125, bends, tees, clamps, rubber-lined hangers	1	set	59.84*	60*	estimate
Section subtotal (excl. “—” items)				658	
Control cabinet CC					
OWEN PR200-24.4.2.0	1	pcs	208.79	209	estimate
OWEN PRM-24.1	1	pcs	90.52	91	estimate
24 V PSU w/ battery charging (Mean Well DRC-100B, 100 W, or equiv.) + 2 batteries 12 V 7 Ah	1	set	131.64*	132*	estimate
Contactorm KM1 2P 25 A 230 V, power controller U1 0–10 V on heatsink, relays K1–K4	1	set	113.69*	114*	estimate
MCBs QF-T, QF-V, QF-PC, QF-24; terminals, rails, cable ducts, busbars	1	set	71.81*	72*	estimate
CC welded enclosure 1000×1060×300: 1.5 mm sheet on a 30×30 tube frame, continuous welds, roof without openings, posts extended 600 mm down, bottom with grille and filter, 2 fans 24 V, powder coating	1	set	203.45*	203*	estimate
IP68 cable glands for Ø16–Ø32 conduit w/ adapters, nuts welded into roof	8	pcs	2.99*	24*	estimate
CC door: 2 tempered glass panes 4 mm 1000×540 with ground edges, brush seal, double-groove guides, roller, wire rope, bracket with quick-release pin	1	set	59.84*	60*	estimate
Linear actuator 24 V, 550 mm stroke, up to 300 N, built-in limit switches + 70×70 cover	1	set	59.84*	60*	estimate
Actuator reversing relays (2 pcs), door button on side	1	set	17.95*	18*	estimate
Section subtotal (excl. “—” items)				983	

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Item	Quantity	Unit	Price, \$	Sum, \$	Price basis
Sensors					
iArduino CO ₂ RS-485 (NDIR + t + humidity) — room, sensor 2	1	pcs	57.44	57	estimate
iArduino climate unit RS-485 (t, humidity, light) — bathroom and cabinet	2	pcs	10.29	21	estimate
iArduino eCO ₂ /TVOC RS-485 (SGP30) — odor/smoke in supply, sensor 9	1	pcs	29.80	30	estimate
R4DCB08 module: 8 channels DS18B20 → RS-485 Modbus	1	pcs	13.16*	13*	estimate
Sealed DS18B20 probe, 2–3 m cable: outdoor and after heater	2	pcs	2.39*	5*	estimate
Smoke detector IP 212-39/1 “Agat”, 4-wire, 7–30 V, relay, no sounder (BD1)	1	pcs	10.77	11	estimate
Differential pressure switch EKF RVG-10-500 (with tubes and fittings): flow SP1, filter SP2	2	pcs	15.56	31	estimate
Cabinet door reed switch SQ1 (IO 102-2 or equiv.)	1	pcs	2.39*	2*	estimate
USB–RS-485 adapter for setting sensor addresses (one-off)	1	pcs	4.79*	5*	estimate
Section subtotal (excl. “—” items)				175	
Computer — for the future, not included in total					
Graphics card GeForce RTX 3060 12 GB (new)	1	pcs	0	0	later
Mini-ITX board with Wi-Fi + CPU + 32 GB + NVMe 1 TB + PSU SFX 550 W + PCIe riser + USB–RS-485	1	set	0	0	later
UPS 650–850 VA (new)	1	pcs	0	0	later
Monitor 15.6” + VESA mount	1	pcs	0	0	later
Section subtotal (excl. “—” items)				0	
Climate and video					
Air conditioner Midea Persona MSAG4-09HRN8-I, black, 2.6 kW	1	pcs	451.06	451	estimate
Midea Wi-Fi module for the AC (local control)	1	pcs	29.92*	30*	estimate
AC install: unit on 2 spacers 40×20, 5 cm off wall (conduits to cabinet behind it), route ≈4.5 m: copper + cable + drain in one 60×40 duct for 2.0 m, one sleeve low in the infill panel, copper above window, drain drips by window; bracket: unit behind glazed sash, bottom ≈30 cm above ground	1	set	—	—	TBC
IP camera 4 MP, 2.8 mm, IR, PoE (CAM1 — bullet, CAM2 — dome)	2	pcs	0	0	later
PoE switch 4+1, DIN rail	1	pcs	0	0	later
LED strip 24 V blue + profile (cabinet backlight)	1	set	14.36*	14*	estimate
Section subtotal (excl. “—” items)				495	
Finishing					
Decor ladder above radiator: 2 posts 40×40, base plates to floor (5 cm from window edges), 4 bent strips 40×20 with sharp kink (top one — handle at 1.70 m), 4 steps 410×160 bent checker plate 3 mm, post tops at 1.76 m turned into wall, base plates on anchors, powder coating	1	set	101.72*	102*	estimate
Section subtotal (excl. “—” items)				102	
TOTAL for all sections (excl. “—” items)				3,812	

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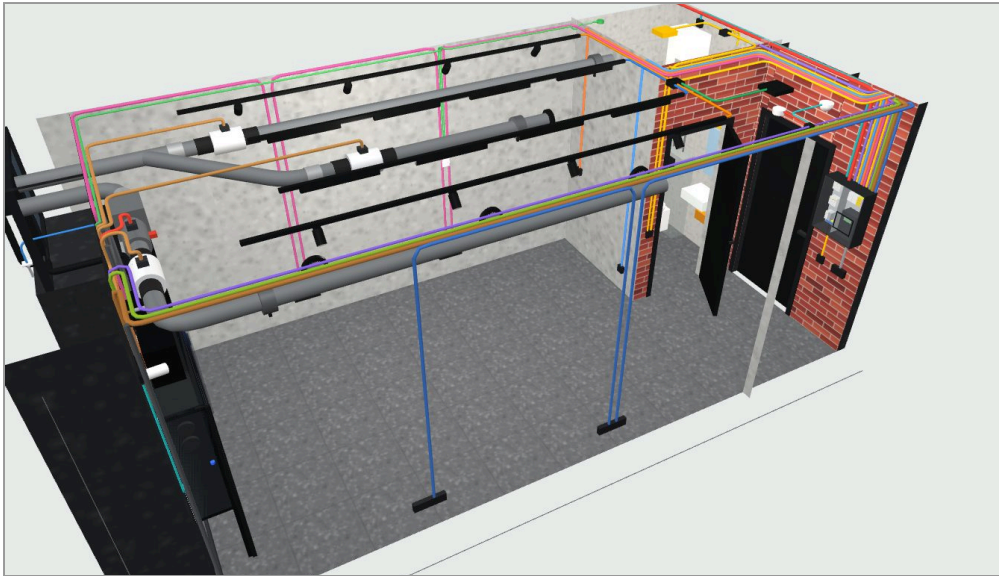
19.6 m² room with bathroom — electrics, lighting, ventilation, automation

Specification (3/3)

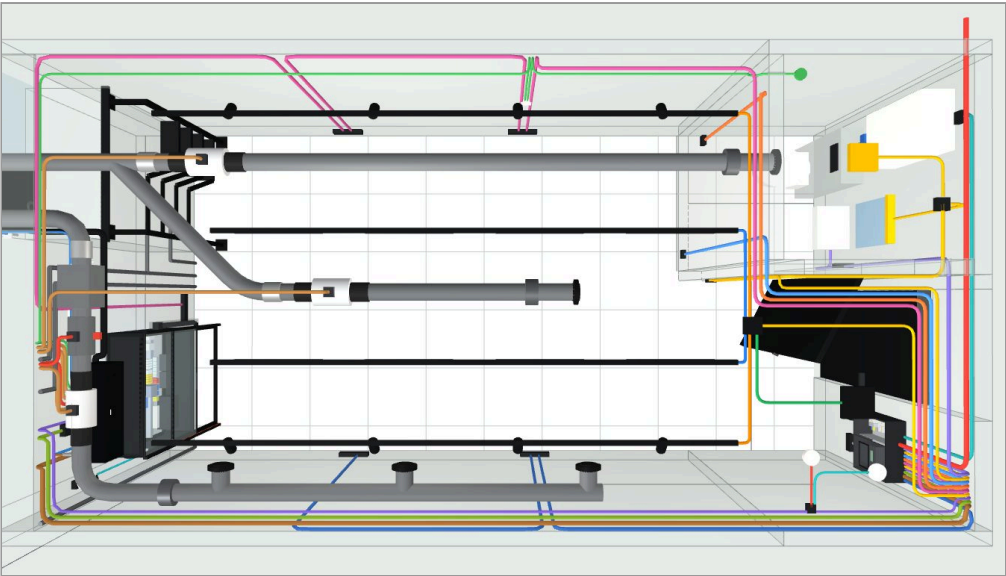
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General view (finish)



Top view (walls transparent)



Control cabinet



Window-to-entrance view

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3D model views

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